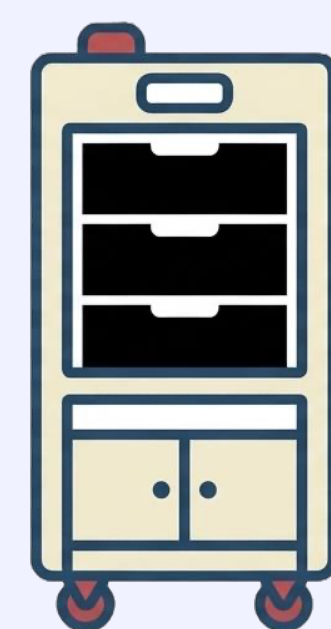




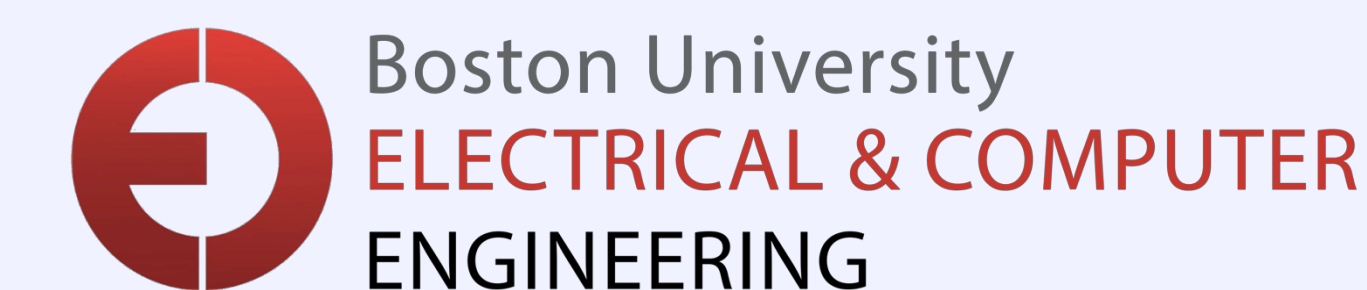
NurseAid

Autonomous Hospital Delivery Robot

Varsha Singh, Kevin Liu, Matthew Kweon, Lillian Chung, Audrey Wang
{varsingh, kliu3816, kweon10, lrchung, axwang@bu.edu}

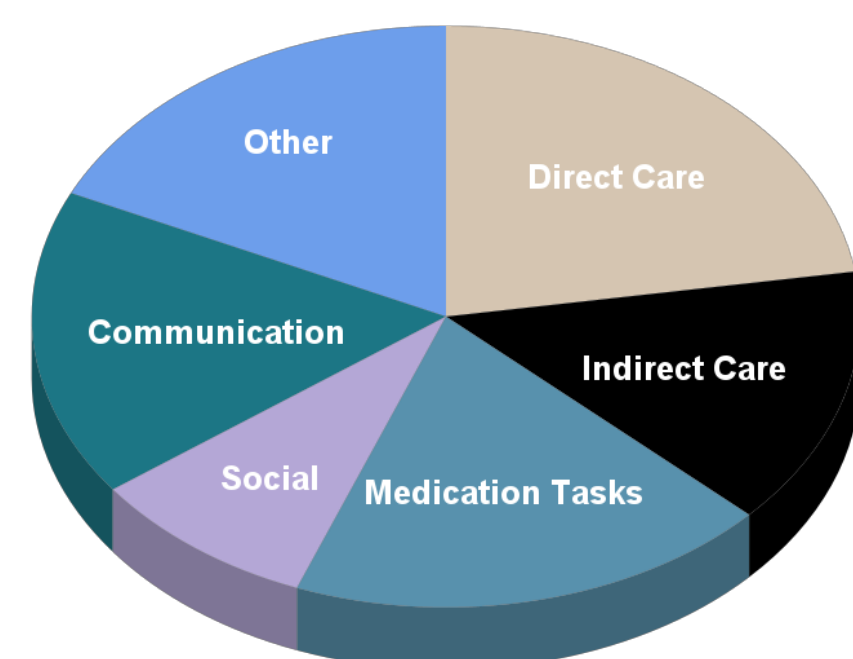


ECE Day, May 1, 2026



Motivation

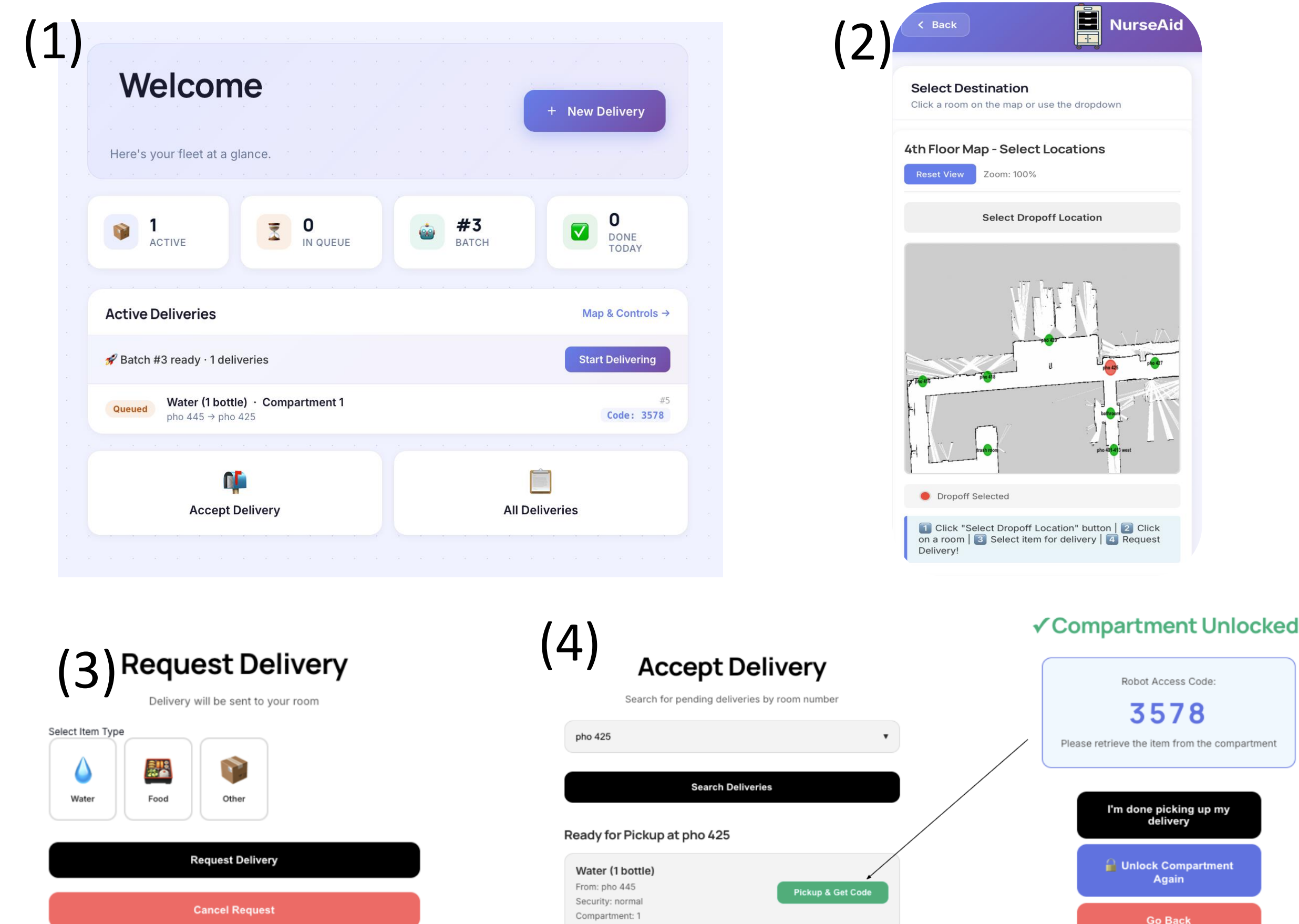
Nurses are burdened with excessive non-clinical tasks, such as delivering food, water, and other supplies, which reduces the time available for direct patient care.



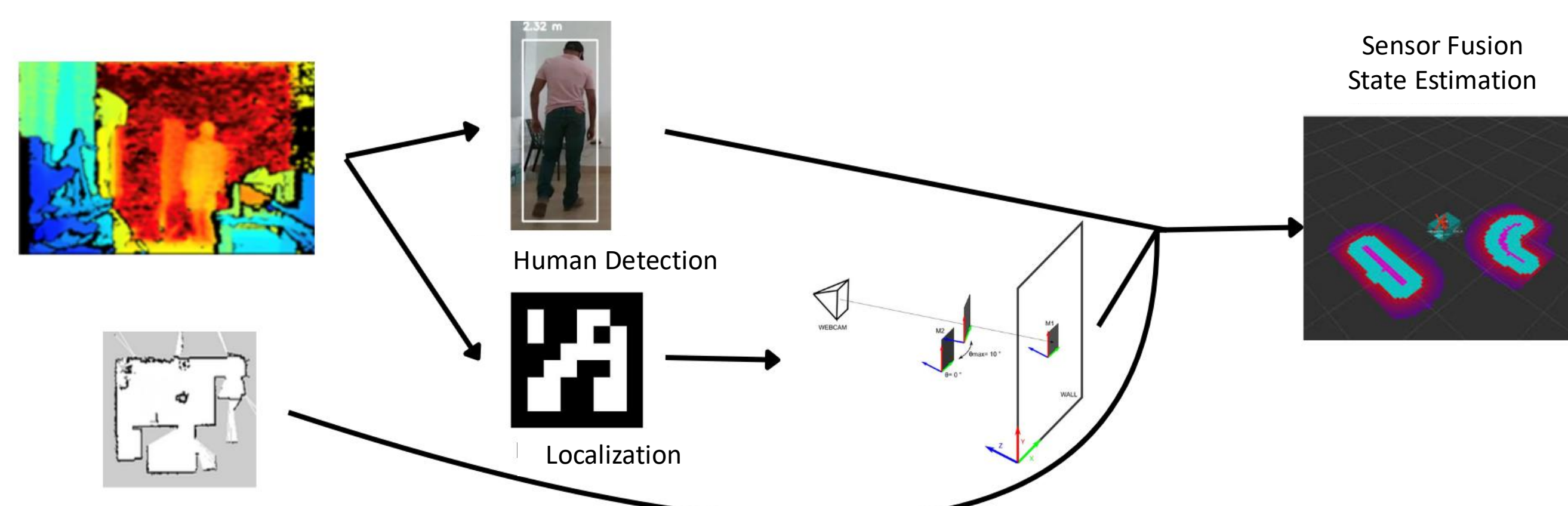
- 1/3 of nurses' time is spent on non-clinical tasks
- Nurses complete an average of 75 tasks per hour
- Patients wait on average 13 minutes for a response to call-light requests

User Experience

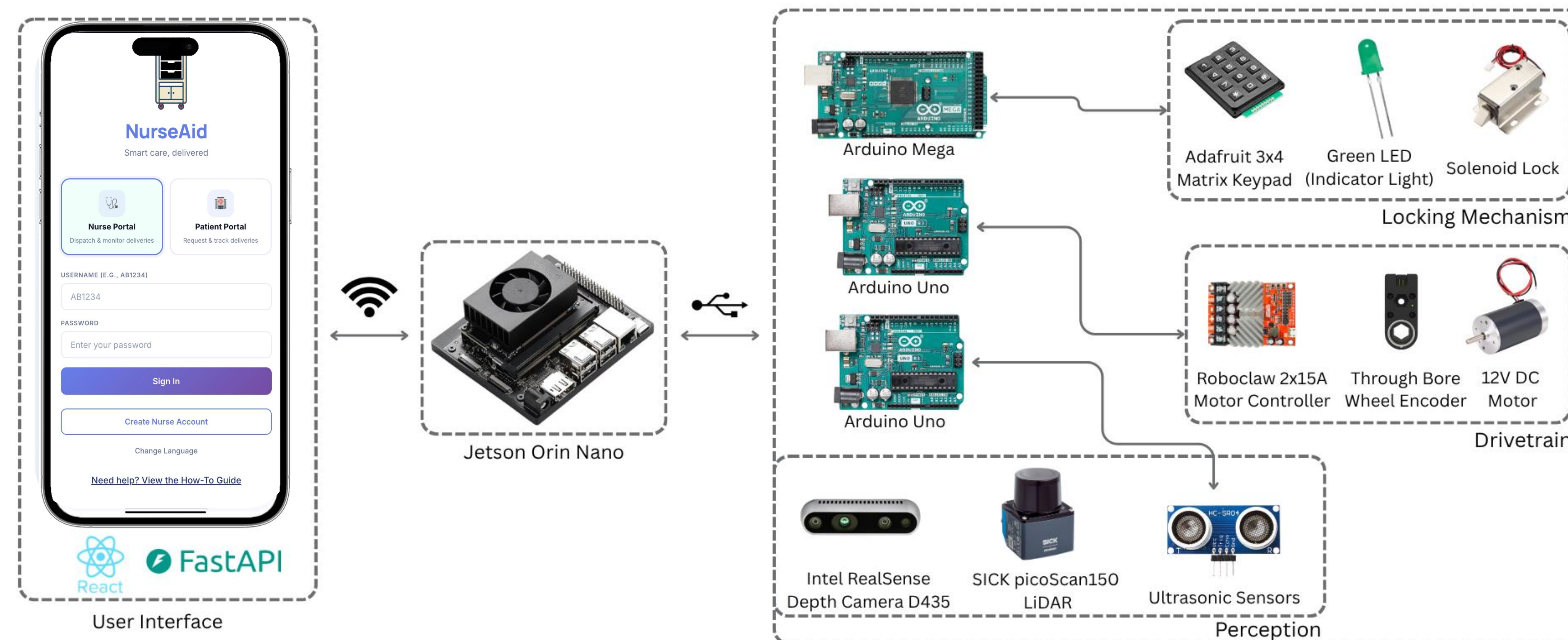
Intuitive Mobile Interface: Abstracting robotics into a simple dispatch suitable for healthcare



Multi-Modal Sensor Fusion



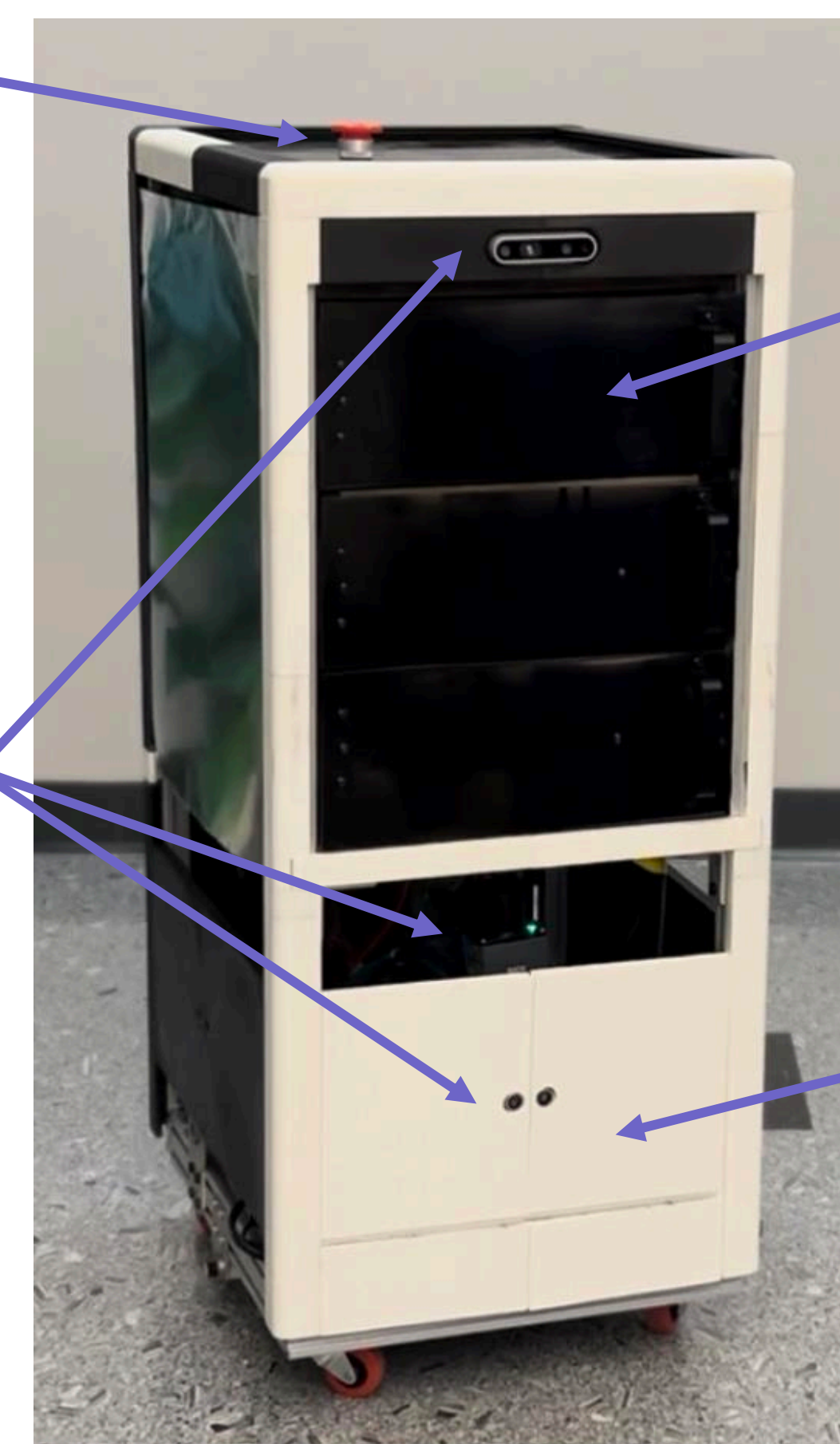
System Architecture



Safety

Emergency Stop Button: Manual override to stop robot motion

Perception Stack: Dynamic obstacle avoidance and human detection

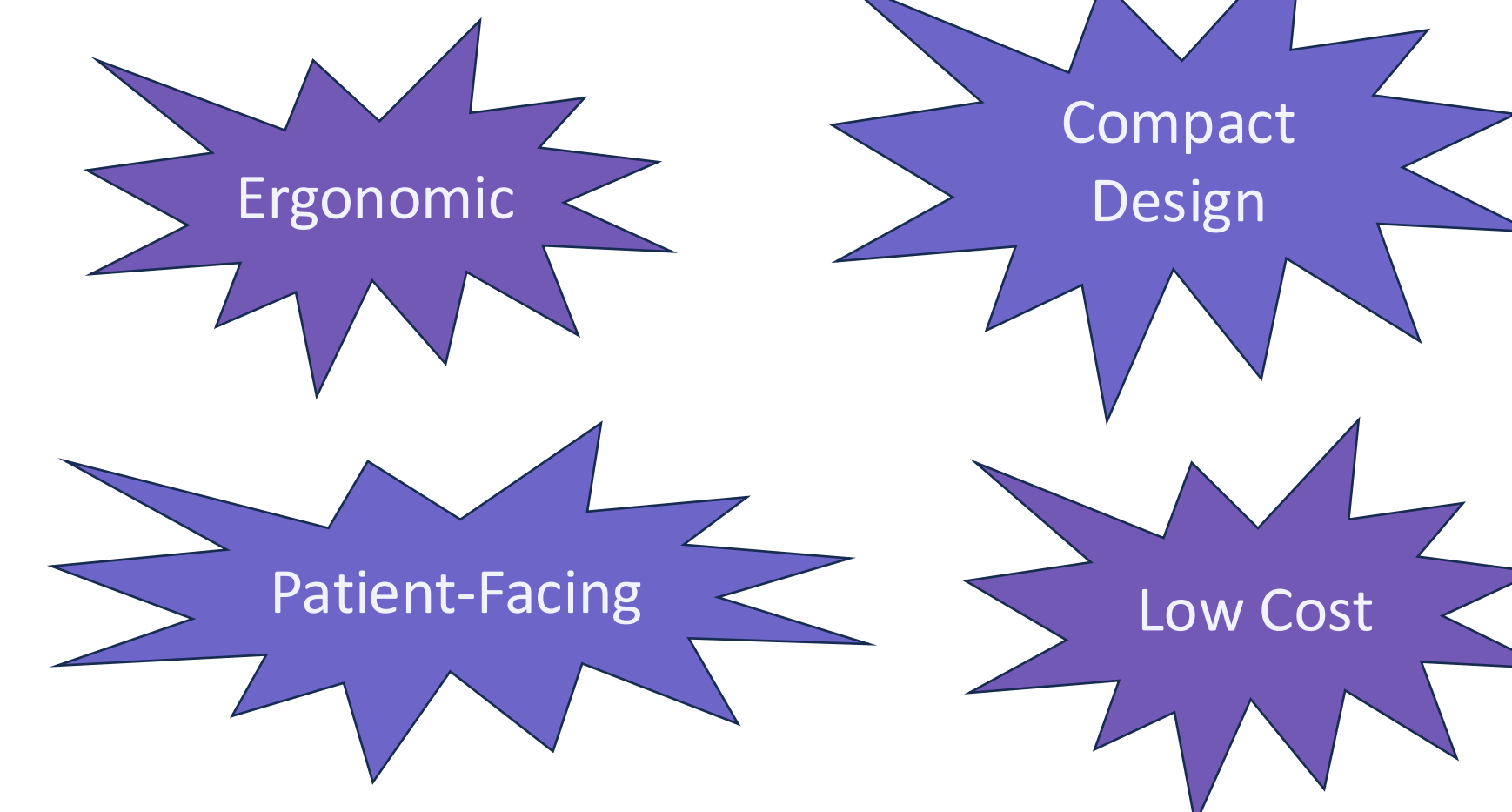


Individually Locked Delivery Compartments: Prevents unauthorized access to contents e.g. medications, lab samples

Outer Shell: Minimizes injury in case of collision

Business Model

Competitive Advantage:



Robotics-as-a-Service Model:

- Option 1: Monthly subscription with maintenance plan
- Option 2: One-time purchase with optional maintenance plan

Acknowledgements

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